Informal Intro to least Squares

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- Minimize sum of squared errors
- More equations than unknowns

Let
- x be our state
- z_i be predicted observation $\quad \hat{z}_i = f(x)$ be actual observation

GOAL estimate x that explains z_i actual observations

Ex: x = 3D features, f: 3D → 2D

We estimate x = real 3D, given 2D observations z_i

ERROR $e_i(x) = z_i - f_i(x)$ = real − predicted

- Gauss dist, mean 0
- Ω info matrix weights of measurements
- $e_i(x) = e_i(x)\,\Omega_i\, e_i(x)$ SQUARED error scalar val

We want $\quad \hat{x}^* = \underset{x}{\operatorname{argmin}} \sum c_i(x)\, \Omega_i\, e_i(x)$

Solving

Iterative Local Linearizations

① Linearize near first guess ② Find derr of error func

③ Set to 0, solve for hopefully better state

○ linearizing

- Taylor Expansion of error around guess

$\rightarrow e_i(x+\Delta x) \approx e_i(x) + \underline{J}(x)\Delta x$

Jacobian since multivar

$$J = \begin{pmatrix} \frac{\partial f_1}{x_1} & \cdots & \frac{\partial f_1}{x_n} \\ \vdots & & \\ \frac{\partial f_n}{x_1} & \cdots & \frac{\partial f_n}{x_n} \end{pmatrix}$$

○ Squared error

$\rightarrow e_i^2(x+\Delta x) \approx (e_i + J_i\Delta x)^T \Omega_i (e_i + J_i\Delta x)$

$\vdots \; = e_i^T \Omega_i e_i + 2e_i^T \Omega_i J_i \Delta x + \Delta x^T J_i^T \Omega_i J_i \Delta x$

$$\underbrace{}_{C_i} \quad \underbrace{}_{b_i^T} \quad \underbrace{}_{H_i}$$

$$= C_i + 2b_i^T \Delta x + \Delta x^T H_i \Delta x$$

b Dir and mag to decrese error

→ How wrong × trust × map error to state var

Now Global Error

$$F(x+\Delta x) = \underbrace{\sum C_i}_{c} + 2\Big(\underbrace{\sum b_i^T}_{b^T}\Big)\Delta x + \Delta x^T \Big(\underbrace{\sum H_i}_{H}\Big)\Delta x$$

$$\left[F(x+\Delta x) \approx c + 2b^T\Delta x + \Delta x^T H \Delta x\right]$$ where $b^T = \sum e_i^T \Omega_i J_i$, $H = \sum J_i^T \Omega J_i$

∴ Minimize This!

1 Find Derr 2 Set to 0 3 Solve

Hessian

how strongly vars constrained

"How hard is it to move that way"

Minimizing

1 IF $f = x^T H x + b^T x$, $\frac{\partial f}{\partial x} = (H + H^T)x + b$

thus we have $\frac{\partial f(x+\Delta x)}{\partial \Delta x} \approx 2b + 2H\Delta x$

2 $0 = 2b + 2H\Delta x$

3 $\underline{H\Delta x = -b}$ → $\Delta x^* = -H^{-1} b$

Gauss Newton we iterate ↑ these steps

Update $x = x + \Delta x$ UNTIL $\Delta x \approx 0$

How to Solve

One Soln: Cholesky Decomposition

IF A pos, semi, $Ax = b$

$\rightarrow$ $A = LL^T$ with L is lower triangular

① Solve $Ly = b$ with $y = L^T x$

② Solve $L^T x = y$

EXAMPLE Odometry data

- u_i measure
- u_i^* real value

We want to find parameters for correction function f such that

$$u_i^* = f(u_i) = \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{pmatrix} u_i$$

- State Vector: params x_{ij}, $x = (x_{11}\ x_{12}\ x \ldots \ldots x_{33})^T$
- Error Func: $e_i(x) = u_i^* - f(x)$

$$= u_i^* - \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{pmatrix} u_i$$

$\rightarrow$ Derrivative $\dfrac{\partial e_i(x)}{\partial x} = J_i = -\begin{pmatrix} \frac{f_1}{\partial x_{11}} & & \\ & \ddots & \\ & & \frac{\partial f_3}{\partial x_{33}} \end{pmatrix}$ where $f_i = i^{th}$ row $\times u_i$ since $(\ldots)u_i$ gives 3×1 vec

$$= -\begin{pmatrix} u_{ix} & u_{iy} & u_{i\theta} & & & \\ & & & u_{ix} & u_{iy} & u_{i\theta} \\ & & & & & u_{ix} \quad u_{iy} \quad u_{i\theta} \end{pmatrix}$$

Qs

- How many Observations we need?
  - $\rightarrow$ 9 vars, 3 per obv, so min 3

- Non Linear Least Squares
- Manifolds

With State Estimation

With x = State u = Controls z = Obs

$$P(x_{0:t} \mid z_{1:t}, u_{1:t}) = \eta P(x_0) \prod_t [P(x_t \mid x_{t-1}, u_t) P(z_t \mid x_t)$$

Take log

$$= C + \log \underbrace{P(x_0)}_{N} + \sum \left[\log \underbrace{P(x_t \mid x_{t-1}, u_t)}_{N} + \log \underbrace{P(z_t \mid x_t)}_{N}\right]$$

Assume all Ps gaussian

$$\cdot \log N(x, \mu, \Sigma) = C - \underbrace{\underbrace{\tfrac{1}{2}(x-\mu)^T}_{e^T(x)} \underbrace{\Sigma^{-1}}_{\Omega} \underbrace{(x-\mu)}_{e(x)}}_{\substack{\text{Square Error} \\ e(x)}}$$

We get $\log P(x_{0:t}; z_{1:t}, u_{1:t}) = C - \frac{1}{2} e_p(x) - \frac{1}{2} \sum [e_{u_t}(x) + e_{z_t}(x)$

Which if we maximize log likelihood, we minimize error

$$\text{argmax} \log P(x_{0:t} \mid z_{1:t}, u_{1:t})$$

$$= \text{argmin}\ e_p(x) + \sum [e_{u_t}(x) + e_{z_t}(x)]$$

$\therefore$ Soln to least square $\approx$ finding mean = most likely probability